Supplement to: PakEmis v1.0: an evidence-graded public-data reconstruction of Pakistan's non-LULUCF greenhouse-gas emissions, FY2015–16 to FY2023–24

The air beta, the parallel-table rule, the reproduction record and the claim matrix

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Release provenance. This report documents one release of PakEmis and is generated against it. Where a sector report and the release's generated product catalogue (`outputs/pakemis_product_catalog.csv`) disagree, the catalogue is authoritative: sector reports are revised on their own cadence and may lag the products.

Release version	`1.0.0rc67`
Commit	`1876f398f85c`
Reporting period	FY2023-24 (2023-07-01 to 2024-06-30)
Baseline configuration	S (survey basis)
National total	444.96 Mt CO_2e (AR5) under S, the reference; 450.20 Mt under L
Release manifest	`13462fb4c2d1ea62` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

This supplement carries what a reader of the main article may need and the article itself does not hold: the air-pollutant beta layer (Section S1), which is deliberately kept out of the greenhouse-gas result; a pointer to the key-category analysis (Section S2); the parallel-table rule in the IPCC defaults (Section S3); the reproduction record, with where the months and the geography come from and the source-by-source account (Section S4); and the claim matrix (Section S5), the generated register of every headline claim the article makes and the release product each one resolves to.

S1 The air beta

This layer is a labelled beta and is not part of the greenhouse-gas result. The layers below are computed from the same activity quantities as their sectors' greenhouse-gas estimates, which is the property worth reporting; they are *not* of comparable assurance to each other or to the greenhouse-gas inventory, and the ordering should be read as a statement about what is estimated rather than about the country's sources. Every figure here is read from the sector output named beside it, and the manuscript-number tests cover this document on the same terms as the main paper.

Since this revision the layer is a hashed part of the release rather than a loose companion: the ten sector tables (`air_pollutants_2024.csv` with `air_pollutants_provincial_2024.csv`, `air_pollutants_district_2024.csv` and `air_pollutants_timeseries_1990_2024.csv` beside it, plus the six per-sector files) carry release digests, and the semantic layer republishes them as five queryable tables — `dim_pollutant` (the fourteen-species union, with the containment that forbids summing across species: BC and OC inside $PM_{2.5}$, $PM_{2.5}$ inside PM_{10}, PM_{10} inside TSP), `fact_air_annual` (the union of the seven sector layers with each layer's assurance word), `fact_air_timeseries`, `fact_air_adm1` and `fact_air_adm2` (agriculture's provincial and district allocations). The query contract's `query_air` serves them in native kilotonnes of the named species, never as CO_2e, and refuses a sum across pollutants rather than computing one.

The cube now also publishes its per-stream district vectors (`pakemis_adm2_stream_reference.csv`): the `stream_shape` accumulation the shipped ADM2 marginals are built from, at the grain downstream layers inherit from — every placed stream closes to its provincial-year mass at 10^{-9} relative, and the only absent streams are aviation's three cruise streams, unallocated below the provinces by construction.

On that reference the beta's whole geography now ships: every sector's pollutant rows carry their own streams' provincial vectors (`air_pollutants_adm1.csv`, with the vector's status and the stream's evidence grade on every row) and, below the provinces, the streams' canonical-frame district vectors (`air_pollutants_adm2.csv`) — agriculture by its own allocator re-keyed to the same frame, aviation's cruise share held at the provinces exactly as its greenhouse-gas mass is, and each layer closing to its national total at the allocator's own tolerance.

S1.1 The nine-fiscal-year covered-sources panel

The beta's single reference year now sits inside a nine-fiscal-year panel (FY2015–16..FY2023–24), built the way the greenhouse-gas national series is built: each sector's own air engine run for each year under the sector's own fiscal convention, never a national total distributed downward. The atoms (`pakemis_air_atomic_annual.csv`, one row per period, sector, category, process and species, in Gg) each carry their activity basis, their factor source and their own source-stream anchor; the rollup (`pakemis_air_annual_fy2015_16_fy2023_24.csv`) declares `coverage_scope=covered_sources_only` on every row — and reaches the query layer as its own semantic table (`fact_air_panel.csv`), every answer over it carrying that scope as a verbatim limitation — because this is not a complete national seven-species inventory and does not claim to be. What is and is not estimated is itself a product: `pakemis_air_coverage_matrix.csv` types all 630 period-by-module-by-species cells under the closed IPCC notation keys — a missing estimate is written as its key, never as a zero. Power's screening scenarios run for the same nine years and ship beside the panel (`power_air_screening_fy2015_16_fy2023_24.csv`), outside it, so a scenario end is never laundered into an estimate; industrial combustion (rc58), IPPU process pollutants and fugitive NMVOC with flaring (rc59) are inside the panel as provisional estimates from one source-class register, and the classes that register cannot price (urea finishing, nitric acid, lime, glass, venting, post-meter NMVOC) stay typed absences in the matrix with their gap-register rows. The reference-year slice of the panel reproduces the single-year layers above to rounding, and the series honesty is carried in the atoms' own bases: agriculture's crops and nitrogen on their fiscal Economic Survey series with livestock on the declared inter-census projection, residential's five untranscribed fuels held and labelled, and the road and rail sulfur regimes stepping at the cited Euro-II and Euro-V import-mandate boundaries. The panel now carries the residential dung sensitivity as a second configuration. The reference, `REF_2023_24`, interpolates the dung series per household between the two HIES rounds and never claims nine observed years (each year's basis is labelled); `L_HERD1624` is configuration L as a nine-year series, all the dung agriculture diverts burned in households, the reference year reproducing that configuration's shipped particulate exactly, and `residential_configurations_by_year.csv` is the register both configurations' dung masses reconcile against.

The panel is spatial for all nine years, on the same 0.05° lattice and through the same dispatch as the greenhouse-gas cube: each pollutant mass rides the cell vector its own anchor stream was placed on for that period, so the grid (`pakemis_air_grid_0p05_annual_fy2015_16_fy2023_24.nc`, with the ledger-anchored configuration as its own file, `pakemis_air_grid_0p05_annual_ledger_fy2015_16_fy2023_24.nc`), the provincial table (`pakemis_air_adm1_by_year.csv`) and the district table (`pakemis_air_adm2_by_year.csv`) are three marginals of one allocation, closing per period, configuration and species against the panel with the cruise share typed in `pakemis_air_unallocated_by_year.csv` rather than zeroed. The by-year tables carry the (period, stream) join key and no evidence badge of their own — whether a year's geography was measured or borrowed is the weight registry's to say, and `pakemis_air_spatial_support_by_year.csv` publishes that join per pollutant mass, with the activity/factor and spatial halves of the uncertainty reported side by side and never combined in `pakemis_air_uncertainty_split_by_year.csv`. The family is servable: the semantic

layer carries it as `fact_air_adm1_year.csv` and `fact_air_adm2_year.csv` (kilotonnes, the products' gigagrams verbatim), and `query_air` answers fiscal-period questions at provincial and district grain under either configuration, with the covered-sources limitation on every envelope.

What the panel's large typed absence plausibly holds is itself a product — the completeness-mass rule applied to air. Power's screening carries BOTH scenario ends (the uncontrolled default and a controlled case crediting every unit with the named held-table control), grid only since rc58, and `pakemis_air_absence_screening_by_year.csv` serves the pair per period and species — scenario ends, not mathematical bounds — beside every fiscal-period air answer (`fact_air_absence_screening` in the envelope's completeness block) and never summed into the covered panel: a covered total plus an uncontrolled end would be a number that is neither an estimate nor a bound. The acquisition that would replace power's pair with an estimate stays named per row (the NEPRA licensee environmental submissions of gap `air-power-controlled`).

Industrial combustion entered the covered panel at rc58 as a *provisional estimate* (`industry_air_pollutants_fy2015_16_fy2023_24.csv`): the same gas, liquid and coal quantities the greenhouse-gas streams book, the cement engine's clinker and the power module's captive gas, each in a source class with its own equipment type and control state, on factors transcribed with page and digest from the EMEP/EEA 2023, AP-42 and South Asian kiln literature now held, and SO_2 from a sulfur balance on dated evidence. Its admission rule is the one the user set — a defensible activity basis, an applicable factor for the actual source class, an explicit process boundary and documented assumptions, with uncertainty — and what cannot meet it is a typed absence with its activity written beside it (NH_3 for every fuel, other-industry coal inside HDIP's cement column, captive solids, industrial bagasse). Captive self-generation is priced once, here, under IPCC 1.A.2, and no longer appears in the power screening. The equipment shares (boilers, furnaces, engines and turbines) are declared rather than surveyed and carry 40–50 percent; the brick kilns carry most of the layer's SO_2 and particulate on Punjab coalfield sulfur ranges rather than a measured average. That gap has been closed. Three as-fired analyses of the coal Pakistani brick kilns actually burn are now held—2.38 percent from a SAARC Energy Centre audit of a Bull's trench kiln at Lohe Bher (2012), and 5.60 and 5.95 percent from the CDPR/IGC kiln study's laboratory analyses of the Balochistan coal its zigzag kiln fires and the two-to-one mixture its fixed-chimney kiln fires (2022)—and their mean, 4.643 %, replaces the resource-range value the layer carried before. *The ground for the change is the basis, not the level*: the superseded value was the geometric middle of two Punjab coalfields' resource ranges, which the register's own vocabulary defines as what the seams hold rather than what was burned, and the level barely moves. This is the first row in the register to use the `measured_avg` basis, declared since rc58 and until now unearned. The old derivation was wrong independently of its value: both measured Punjab kilns fire Balochistan and Sindh coal bought at differing prices, so the sulfur a kiln fires is an economic choice before it is a geological one. What is not claimed is precision—three batches are not a fleet, every source says so, and the uncertainty was deliberately not narrowed to match the new basis.

A second sulfur value moved with it. Industrial and main-activity power furnace oil were booked at 3.0 % and 1.3229 %: one national fuel, the same years, a factor of 2.27 apart, each row carrying a note explaining why the other was wrong. The furnace-oil import grade split transcribed from PBS Compendium Table B-10 settles it—the pool runs 85.3 % to 100 % high-sulphur grade across the four years the table covers, so it sits near the top of the 0.5 % to 3.5 % specification span rather than at its unweighted middle. The power row is retired, with its reasoning preserved, and the power classes now cite the same row as industry, so the disagreement cannot be restated by a later edit. Equipment continues to change the *retention*, which is declared per class.

Together these raise Lahore's FY2015–16 SO_2 to 26.49 Gg, close to a third above what the district carried before, and steepen its nine-year *endpoint change*—the ratio of the closing year to the opening one, not a fitted rate—by about four points. Stream by stream and year by year, with every sign shown, the revision is published in `pakemis_sulfur_adoption_effect.csv`; every row it moved is an *increase*. The mechanism is not the one a reader would expect: *neither stream's own path changed.* Industrial SO_2 falls by the same fraction before and after, and so does power. The aggregate moved because power's share of the opening year rose to 40 % from roughly half that, and power declines

faster than the district does. A sulfur value is a flat multiplier within a stream and cannot bend that stream's path; re-weighting streams whose paths differ is what moves an aggregate. What the correction supplies is a better physical assumption— not a new observation of how fuel or sulfur moved through time.

Three cautions belong with that number, and the first two were overstated in an earlier draft of this section. The district's fall is *not* dominated by any single documented policy. Its transport component is almost entirely the modelled diesel sulfur step, which is declared from import-mandate coverage rather than measured at the pump; road diesel consumption *rose* over the step year, and the step is smaller than a tenfold sulfur cut would give because petrol's declared sulfur never moved. Its industrial component is not synonymous with brick kilns, which is why the effect table is cut per source class. And the brick column that dominates that component is a residual by its publisher's own footnote—estimated by deducting other uses from total production—whose value in the closing year rests on a supply table the publisher flags as partial and which the next edition revised, with the supply side revised in lockstep. Where that column exceeds the domestic non-Thar coal available, the record types the breach *conditional* and names its conditions: neither coal imports by consumer nor stock change is held, and the publisher's own footnotes place imported coal in the adjacent column. Imports in those years exceed domestic production. The breach establishes that the reported quantity is incompatible with a domestic-only supply assumption; it does not establish which side is wrong.

A fourth caution is asymmetric and works against the published figure. Lahore's power SO_2 is one station, and its opening year is priced from generation through a heat rate while its closing year is priced from a metered fuel quantity. Where both routes exist the generation route runs high, so the opening year is likely overstated relative to the close and the endpoint change correspondingly overstated—by roughly a point, on the bias measured in `pakemis_air_lahore_power_register.csv`. That correction is not booked, because a bias measured on one station in five years is not established for the four earlier ones; it is recorded so the asymmetry is visible rather than implicit.

Retention remains unmeasured, and the reason is now a hard one rather than an absence. The only study pairing fuel sulfur with stack SO_2 on Pakistani kilns publishes, for one of its two kilns, a fuel-mass emission factor 2.5 times what that kiln's own fuel sulfur can supply, and its carbon fails by a wider margin still; its two published factors reproduce each other through the calorific value, so the fault lies upstream in that kiln's fuel-rate or stack-flow term. The declared 0.25 retention therefore stands unchanged, and it is a single term standing where fuel retention, raw-material sulfur and any installed control should be distinguished. Every verdict above, including three readings of the brick test that were withdrawn before the fourth was adopted, is recorded in `pakemis_fuel_sulfur_decision.csv`. What each sulfur row was measured on, and how far it has been carried from that population, is stated per row in `fuel_sulfur_evidence.csv`: the brick central is three batches from three kilns on two dates, applied to every kiln in the country in all nine years, and the register now says so in its own columns rather than leaving `measured_avg` to imply otherwise.

The same allocation the grid and provincial tables are marginals of is cut for Lahore District (`pakemis_air_lahore_by_year.csv`), every row beside its anchor streams' weight-registry basis and the national mass no geometry placed below the province.

S1.2 Assurance, layer by layer

S1.3 Magnitudes

Within that caveat the contrast is stark. Residential solid fuel is 5.4 % of national greenhouse-gas emissions and emits 282.7 kt of primary $PM_{2.5}$ and 20.0 kt of black carbon under the reference `REF_2023_24` (631.9 and 41.5 kt under configuration L)—more than the other estimated sources combined. Road transport is 1.7 times its size in CO_2e, at 9.0 % of the national total against residential's 5.4 %, and emits 8.2 kt of primary $PM_{2.5}$: about a thirty-fourth as much, on the EMEP/EEA Tier 1 factors that replaced the earlier uncited per-kilometre diagnostic, with evaporative and wear processes now inside the layer. Open waste burning contributes 63.9 kt and crop-residue burning 54.0 kt, both far above transport despite being marginal in CO_2e. Household biomass likewise dominates carbon monoxide, at 3353 kt against transport's 1430 kt.

Layer	Status	What limits it
Residential solid fuel	estimated	stove-level factors are declared judgement
Agriculture (burning)	estimated	burned mass bottom-up; timing not resolved
Waste open burning	estimated	burned fraction is a declared parameter
Railways	estimated	one fuel, one factor family
Road transport	Tier 1	fleet-average factors (EMEP/EEA Tier 1); no technology vintage resolution
Power	uncontrolled bound	no control-device inventory obtained
Aviation	LTO screening	airport cycles only, not a full airport inventory
Industry	provisional estimate	source-class typical controls; declared equipment shares; SO_2 by sulfur balance on dated evidence
IPPU	provisional estimate	ammonia, electric-route steel and soda ash on EMEP Tier 1/2 process factors; urea, nitric acid, lime and glass typed absent
Fugitives	provisional estimate	NMVOC per segment on the module's own volumes; refining, refinery flaring and satellite-observed field flaring; venting and post-meter typed absent

Table S1: Assurance class for each air-pollutant layer. Two of the ten are bounds or screening estimates and three are provisional estimates from the source-class register (industrial combustion on declared equipment shares; IPPU process pollutants and fugitive NMVOC with flaring on the modules' own activity), each with named classes it cannot price, so what follows orders what is estimated rather than the country's sources. The classes are declared in `pkinv.report.figures.AIR_ASSURANCE` and a test asserts this table prints them.

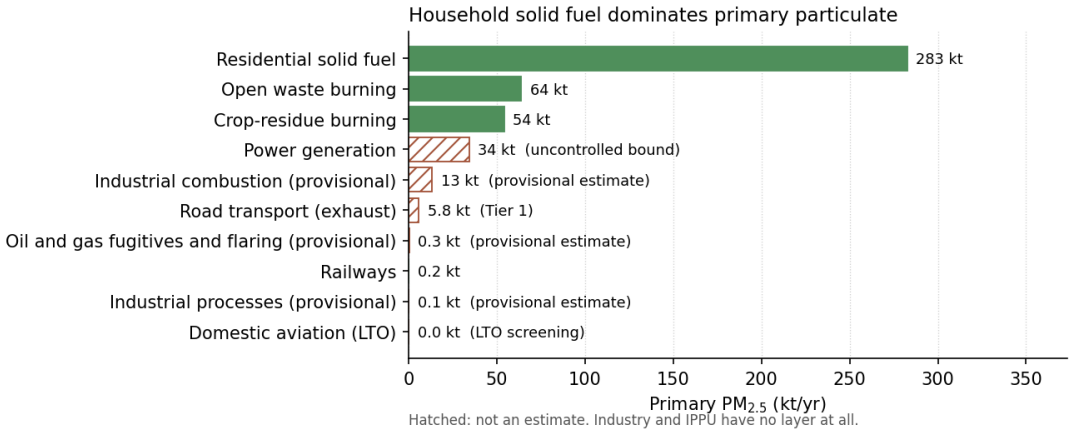


Figure S1: Primary $PM_{2.5}$ by combustion source, for the sectors carrying an air-pollutant layer. Residential solid fuel emits roughly 34 times road transport's primary $PM_{2.5}$ under `REF_2023_24` and 77 times under configuration L, while contributing less than half again its CO_2e. Hatched bars are the bound and the screening estimate and should not be ranked against an estimate. Each layer is computed from the same activity quantities as its sector's greenhouse-gas estimate, so the ordering is a property of combustion conditions rather than of separate studies with different activity bases.

S1.4 What the comparison does and does not support

The policy reading is that the two inventories rank sources almost inversely. A mitigation programme selected on CO_2e would target transport and electricity; a programme selected on primary $PM_{2.5}$ emitted would target household cooking and heating first by a wide margin, followed by two open-burning categories that are together 1.0 % of the national inventory. Because each estimate derives from its own sector's activity quantities rather than from a separate study, that divergence is a real feature of how these fuels burn rather than an artefact of mismatched activity bases—which is the usual reason such comparisons are unreliable in developing-country contexts.

Three limits bound how far it can be taken. *Emitted mass is not exposure*: the burden a population receives depends on where and how each source releases, which this inventory does not model, so the ordering identifies where the particulate mass originates and not where the harm lands. *The layer is partial*: the power figures are uncontrolled bounds rather than what the fleet emits, industrial combustion is a provisional estimate whose equipment shares are declared rather than surveyed, IPPU process pollutants and fugitive NMVOC are provisional estimates with urea finishing, nitric acid, lime, glass, venting and post-meter losses typed absent, and transport omits cold-start, evaporative and non-exhaust processes. And *the assurance is uneven*, per Table S1, which is why this note is labelled beta and why the main paper reports the one-activity-basis finding without the ranking.

S2 Key categories

The key-category analysis—level against variance, the two rankings it produces and what the ten-sector trend window can and cannot say—is reported in full in the main article's Technical validation section (Section 6.7) and is not repeated here: a fragment is printed in one document only, so that the number tests hold each printed value in exactly one place. This slot is retained so that the supplement's numbering follows the outline of the data-description article.

S3 The Parallel-Table Trap in the IPCC Defaults

The 2006 Guidelines (IPCC, 2006) publish stationary and mobile combustion factors not as one table per fuel but as several parallel tables, one per source category. Volume 2 Chapter 2 gives four for stationary combustion—energy industries, manufacturing, commercial, and residential—and Chapter 3 gives separate tables for road, rail and off-road mobile sources. The same fuel appears in all of them.

Across those tables the CO_2 factor for a given fuel is *identical*, because CO_2 depends only on the carbon in the fuel and on complete oxidation. The CH_4 and N_2O factors are not identical, and in some cases differ by nearly an order of magnitude, because they depend on how completely and at what temperature the fuel burns.

This creates a failure mode with an unusually bad signature. A compiler who selects the wrong table gets CO_2 exactly right—and CO_2 is typically 95 % to 99 % of a fossil combustion category, so every obvious check passes. The total looks correct. The comparison against an external CO_2 dataset agrees. Only the trace gases are wrong, and they can be wrong by a factor large enough to move the affected category by 8 %—which is smaller than most published uncertainty bands and therefore invisible to any validation that works at the level of the total.

How much of the total moves depends on how CO_2-dominated the category is, and an earlier edition of this paper overstated it. The railway and mixed off-road cases swap N_2O between 28.6 and 3.9 kg TJ^{-1} and move their own category by 8.0 %. The power and residential cases swap CH_4 on gas streams whose totals are almost entirely CO_2, and move the sector by 0.08 and 0.12 % respectively—the CH_4 row moves several-fold, the total barely at all. This paper previously reported all four as 8 % to 9 % sector errors. The magnitudes are now computed from the engine's activity and the two published factors by `tests/test_factor_table_trap.py`, so the claim cannot be restated without the arithmetic agreeing.

The finding is unharmed by the correction, and is arguably sharper for it. What makes these errors dangerous is not how much of a total they move but that they move NONE of the CO_2: every external comparator this inventory has is CO_2-dominated, so all four are invisible to all of them regardless of size.

We encountered this trap four times in ten sectors. Table S2 gives all four instances with the quantified consequence of the plausible error. Each is pinned by a test that fails if the wrong table's value is ever substituted, and each sector's module carries the warning in its own documentation with a pointer to its mirror image.

Sector	Correct table	Plausible error	Factor difference	Consequence of the error
Electricity generation (1A1a)	Vol 2 Ch 2 Table 2.2, Energy Industries	Table 2.5, Residential	Natural gas CH_4 $1\,kg\,TJ^{-1}$ vs 5	CO_2 unchanged; CH_4 inflated fivefold. A utility boiler burns far more completely than a domestic appliance
Residential fuel (1A4b)	Vol 2 Ch 2 Table 2.5, Residential	Table 2.2, Energy Industries	Natural gas CH_4 $5\,kg\,TJ^{-1}$ vs 1; kerosene 10 vs 3	The mirror image: CH_4 understated by 80 % on a quarter of household energy input
Railways (1A3c)	Vol 2 Ch 3 Table 3.4.1, Railways	Table 3.2.2, Road diesel	N_2O $28.6\,kg\,TJ^{-1}$ vs 3.9—7.3 times higher	Railway N_2O understated by 86 % and the sector total by about 9 %. Large slow-speed locomotive engines run lean at high load, the N_2O-forming regime
Off-road machinery (1A4c)	Vol 2 Ch 3 Table 3.3.1, Off-road	Table 3.2.2, Road diesel	N_2O $28.6\,kg\,TJ^{-1}$ vs 3.9	The category would report about 8 % less. Identical to the rail value because both derive from EEA 2005 and neither engine class generally carries a catalyst

Table S2: Four instances of the parallel-table trap in this inventory. In every case the plausible wrong table leaves CO_2 exactly correct—so the sector's dominant gas, its external benchmark comparison and its mass balance all still agree—while the trace gases move. How far the total moves depends on how CO_2-dominated it is: the railway and off-road swaps move their own *category* by 8.0 %, the power and residential swaps move the CH_4 row several-fold and the *sector total* by 0.06 and 0.12 %. Each magnitude is computed from the engine by a regression test rather than quoted.

S3.1 Why this is a compiler-facing finding rather than a coding note

Three properties make this worth reporting rather than fixing quietly.

It is *systematic*: it arose in four of the six fuel-burning sectors, and in each case the wrong table was the one a reasonable person would reach for. The power sector's fuel is delivered to buildings; the residential sector's largest fossil fuel is natural gas, whose most familiar factor is the utility one; rail and off-road both burn ordinary road diesel bought from the same national supply.

It is *invisible to validation*. We compare each sector against EDGAR, FAOSTAT, Climate TRACE and, where available, Pakistan's own BUR-1. Every one of those comparators is dominated by CO_2 for fossil categories. A sector using the wrong trace-gas table passes all of them. The only thing that catches it is reading the guidelines carefully enough to notice that there is more than one table, which is precisely the step under time pressure.

It is *directional*. The errors do not average out. Three of the four instances would understate the trace gases, because the road and energy-industry tables carry the lower CH_4 and N_2O values and are the more familiar tables. An inventory community making this error at random rates would be biased low on non-CO_2 combustion emissions.

Two related decisions in the same family are worth recording because they run the other way. First, we apply *no separate oxidation factor* anywhere in fossil combustion: the 2006 Guidelines assume complete oxidation and fold it into the emission factor, with Table 1.4 carrying an explicit oxidation column of 1.0 for every fuel, so applying the 1996-vintage 0.98 on top would double-discount. Second, and deliberately contrasting, the waste sector *does* carry an oxidation factor of 0.71, because open burning genuinely is incomplete. Both are pinned by tests, so that neither can be "corrected" into the other.

Finally, the power sector's activity is handled on an energy basis rather than a mass basis for a reason from the same family. IPCC's bituminous and lignite CO_2 factors differ by only 6.8 % per terajoule but by a factor of two per tonne of coal, because their net calorific values differ by 54 % and

lignite's own published range spans 5.50 TJ Gg^{-1} to 21.6 TJ Gg^{-1}. Working in energy units makes that uncertainty disappear rather than requiring an assumption about which coal Pakistan burns.

S4 Reproduction, months, geography and sources

S4.1 Reproduction

The work is split across two repositories along the line this paper draws. This one holds the inventory: the `pkinv` Python package, the activity data it reads, the generated outputs—including the national 0.05° monthly cube and its ADM2 view (Section 4.7)—the tests, this manuscript and the nine sector technical reports, together with the government source documents that could be retained: `data/source_documents.csv` classifies every activity file as archived, derived or not-archived, and 12 of 217 cite at least one document this project used and does not hold (the provenance audit in Section 4). The companion repository holds agriculture *research* built on top of it—the satellite nowcasting system that estimates emissions between census years (Ahmad, 2026b), the machine-learned activity weights and the agricultural inventory manuscript (Ahmad, 2026a)—as the `pkag` package, which depends on this one. The gridded surfaces this release distributes are built and tested here, not there.

The agriculture *sector* is computed here rather than there, because it is 50 % of the national total and because the residential sector's dung-cake closure (Section 4.8.1) reads its nitrogen flow directly. There is deliberately no second copy of the activity data in the companion repository: two copies would drift, which is the failure the closures in Section 4.8 exist to prevent.

Reproducing the inventory requires Python 3.11 and no compiled dependencies for the core computation. Every sector exposes a runner:

```
python3 -m pkinv.sectors.<sector>.run_<sector>
```

for nine of the ten sectors; agriculture's runner is `python3 -m pkinv.sectors.agriculture.run_inventory`, the name it carried before the sector joined the common build. The cross-sector layer is:

```
python3 -m pkinv.report.national_rollup
python3 -m pkinv.report.uncertainty_all
python3 -m pkinv.report.key_categories_all
python3 -m pkinv.report.completeness
python3 -m pkinv.report.figures
```

The full test suite—3882 defined test functions including the 32 contract invariants, the provenance checks, the manuscript number checks and the figure-currency checks—runs with:

```
python3 -m unittest discover tests
```

All manuscripts compile with `make papers`, which uses `tectonic`. Continuous integration, on each pushed commit, installs the locked environment, runs every sector runner and the cross-sector layer, runs the test suite, and compiles the national manuscript and one sector report (railways); the other sector reports, the figure regeneration and the validation package are the release sequence's job, not CI's, and the release records that sequence in `outputs/pakemis_test_run.json` and the package's own logs.

What that does and does not guarantee. Defined numerical outputs, registered table cells, figures, schemas, source references and structural invariants are guarded: a commit in which any of those disagrees with the engine fails the suite, in CI for pushed commits and in the release gate for every content commit. *Scientific interpretation and semantic claims are not guaranteed by the same mechanism.* This paper's own history is the evidence: a gas decomposition stayed stale for three releases inside a declared-debt list that tracked it without verifying it; comparative language such

as "doubles" survived after the ratio it described had changed; and boundary descriptions drifted between documents. Each of those classes now has a test—the debt list is gone, comparative claims resolve through a claims registry, and cross-document boundary statements are checked—but the honest statement is that the guarantee covers what has been registered, and registration is a choice someone makes.

S4.2 Where the months come from

Table S3 reports the temporal-evidence class of every module's emissions, as a share of that module's own total. It is read from `outputs/temporal_profile.csv`, which the national run writes beside the monthly layer itself. The largest reconstructed basis is now electricity generation: NEPRA's Detail-of-Generation listings carry plant-month generation for the reference year, putting grid coal on TE2 and gas on TE3 (the mixed-labelled combined-cycle group is assigned to gas by the annual control's own arithmetic, and an inferred fuel is never graded TE2). The four sectors that anchor on OCAC's monthly customer-class returns—transport, railways, aviation and industrial combustion—carry the other substantial reconstructed block, and the TE1 share within them is the single month of each fiscal year whose cumulative column is the month itself. Residential is wholly TE3: a degree-day model is a profile built from evidence about the process rather than a record of these months. Cement was TE3 beside it on a borrowed calendar-2022 production shape, and is now wholly TE2 - all nine fiscal editions of PBS Table 2 are parsed, so each year carries its own monthly output. It stops short of TE1 because cement OUTPUT is not clinker CALCINATION, however closely the two track.

Table S3: Temporal evidence by module, as a percentage of each module's own emissions. TE1 observed monthly, TE2 reconstructed from monthly administrative records, TE3 process-based profile, TE4 flat and declared. Rows sum to 100 by construction.

Module	TE1	TE2	TE3	TE4
Agriculture	0.0	7.6	8.0	84.4
Electricity generation	0.8	48.9	38.2	12.2
Road transport	8.4	88.5	0.0	3.1
Residential fuel	0.0	0.0	100.0	0.0
Industrial combustion	1.3	13.7	72.7	12.3
Cement (IPPU)	0.0	87.0	0.0	13.0
Waste	0.0	0.0	24.5	75.5
Fugitive oil and gas	0.0	43.0	0.0	57.0
Railways	6.9	93.1	0.0	0.0
Domestic aviation	8.0	92.0	0.0	0.0

S4.3 Where the geography comes from

Table S4 is the spatial counterpart of Table S3: the spatial-evidence class of every module's emissions, as a share of that module's own total, read from `outputs/spatial_profile.csv`. **The class is now derived rather than declared**, and that is this release's second spatial change. It was a literal on each sector engine, checked against nothing at the point it was written; it is now read from two measured facts—what the cube's own weight distributes mass by inside each geometry (`within_province_placement`, one row per geometry set, held against the cube's dispatch by test) and how many of that set's members that can emit carry a coordinate. Every stream in this inventory is controlled provincially, because that is the level the activity statistics exist at, so the class is not about what binds the total; it is about how mass is placed *below* the province, which is the only part an allocation invents. The ledger publishes the declared letter and the derived one side by side and the build fails where they disagree.

Deriving it moved five sets of streams, in both directions. S1 is native source geometry consuming observed per-object activity, and domestic aviation is the only module that earns it: thirty-six of thirty-seven airports located, weighted by PCAA movements per airport. It was S2 on the reasoning

that the LTO cycle's stages are unresolved—a within-airport question this vocabulary does not ask. S2 is geometry the allocation opens without observed per-object activity at full coverage: railways, whose track is native geometry but whose corridor service tiers are declared judgement; captive power, where 1068 of 1105 registered plants are geocoded and 864 carry a gas allocation to weight them by; and, since this release, the grid fleet at its own plant-ledger fuel and the cement works at their company's census clinker—partial-coverage point allocations whose fallbacks are declared ledger events. S3, 71.79 % of the inventory, is an administrative statistic: provincial fuel sales, registrations, census counts, downscaled on declared proxies. The remaining 0.05 % is NA—not a weaker placement but no sub-provincial placement at all.

A third axis, and the one a reader is most likely to need and least likely to ask for: whether the place a stream is put is where it is *emitted*. 82.09 % of the inventory is `physical`. 8.42 % is `origin` — the waste streams, placed at the people who generate the waste, while the methane leaves landfills, drains and burn sites for which no allocation-grade geometry is held. 9.50 % is `accounting` — road, off-road and industrial fuel, booked by OCAC and OGRA to the province of *sale* and burnt wherever the vehicle or the boiler is. So 79.71 Mt, 17.91 % of the national total, is placed somewhere other than where it is emitted, and until this release every product added the three together and returned one number. `pakemis_adm2_monthly.csv` now carries the view as a dimension, so no row mixes them and summing across them is a deliberate act; `spatial_view_profile.csv` publishes the split. This axis is declared rather than derived, because it is a property of what an activity statistic means and no register holds that; what is enforced instead is that a stream which is not physical must name the release geography it does not have.

The largest class move when the derivation landed was the grid power fleet from S2 to S3 — a located fleet the allocation ignored was not a facility proxy—and the largest move since is the same fleet back to S2, EARNED this time: the cube now routes each grid fuel stream through its corroborated stations at their own plant-ledger fuel, with the built-up fallback a declared event per province, and the cement register follows at its company-grained census clinker. The two LNG terminals stay unlocated as a typed refusal: no held source corroborates a coordinate, and a berth anyone can find on a map is still a coordinate this register cannot defend.

The classes describe *allocation evidence*, not coverage. A second, orthogonal axis records whether any geometry exists at all: 3 cruise-aviation streams (0.06 % of the national total) carry the geometry class `S4:no-native-geometry`, meaning a provincial activity control with nothing beneath it. They are *not* in the gridded cube. Industry carried this class through one release candidate - a provincial control and a promise - until rc47 placed it on its own registers (kiln points, the clinker register, built-up); what remains is fuel burned en route, published in `pakemis_spatial_unallocated.csv`. The grid total is therefore strictly less than the national total, by exactly that 0.05 %, and the identity a reader reconstructs is `ADM1 == ADM2 + unallocated` rather than `ADM1 == ADM2`.

S4.4 Sources

Activity data are drawn from Pakistani national and provincial statistics: the Pakistan Statistical Year Book and the 2023 Population and Housing Census (Pakistan Bureau of Statistics, 2023) from the Pakistan Bureau of Statistics; NEPRA's State of Industry Report for electricity fuel consumption; OGRA's petroleum statistics for road, residential and aviation fuel; Pakistan Railways Year Books for rail traction fuel; the Punjab Statistical Bureau's provincial statistics for cement production; the Economic Survey of Pakistan (Ministry of Finance, 2024) and the Ministry of National Food Security and Research (Ministry of National Food Security and Research, 2023) for agricultural activity; and the National Fertilizer Development Centre (National Fertilizer Development Centre, 2024) for fertiliser. The two agricultural citations name the FY2023–24 editions this release reads and archives, not the FY2024–25 editions published since (Ministry of Finance, 2025; National Fertilizer Development Centre, 2025). International sources are used where no national equivalent exists and are flagged as such in the provenance table. FAOSTAT (FAO, 2025) is *not* among the principal ones any more: household firewood is now the HIES survey's own measured consumption, and FAOSTAT survives as a structural comparison against it rather than as the activity basis.

Table S4: Spatial evidence by module, as a percentage of each module's own emissions. S1 native source geometry, S2 facility or network proxy, S3 administrative statistic on a declared downscaling proxy. **NA is not a fourth quality of placement—it is the absence of one.** The S1–S3 scale grades how mass is placed *below the province*, so a stream that is controlled at ADM1 and retained whole rather than spread has nothing on that scale to be. Industrial combustion read 100 % S3 until this release, asserting an administrative allocation beneath the province that does not exist; its 30 streams briefly read NA before rc47 Wave 4 placed them on their own registers; the NA set is now the 3 cruise-aviation streams in `pakemis_spatial_unallocated.csv`. Those streams still carry a province—what they lack is a level beneath it—and the decomposition's `placement_status` and `finest_placement_level` columns say so, which keeps evidence quality and product availability in separate fields. Rows sum to 100 by construction.

Module	S1	S2	S3	NA
Agriculture	0.0	0.0	100.0	0.0
Electricity generation	87.78	12.22	0.0	0.0
Road transport	0.0	0.0	100.0	0.0
Residential fuel	0.0	0.0	100.0	0.0
Industrial combustion	0.0	60.1	39.9	0.0
Cement (IPPU)	0.0	85.4	14.6	0.0
Waste	0.0	89.8	10.2	0.0
Fugitive oil and gas	43.0	0.5	56.5	0.0
Railways	0.0	100.0	0.0	0.0
Domestic aviation	16.5	0.0	0.0	83.5
National	**11.27**	**16.88**	**71.79**	**0.06**

Emission factors are IPCC 2006 Guidelines defaults (IPCC, 2006) with the 2019 Refinement (IPCC, 2019) applied where it supersedes them, except in the agriculture sector's Tier 2 elements (Ahmad, 2026a). Every factor in the repository cites its source table, and a test asserts that no factor lacks a citation and that any informally sourced value is explicitly declared as such.

S5 The claim matrix

Every headline claim of the national paper is a row of `data/claim_matrix.csv`: the sentence as printed, its quantity references into the claims machinery, the source pointer, the stored query, the configuration, unit and rounding, the evidence class, the limitation paragraph it is bounded by, the technical notes it rests on and the artefact that shows it. `scripts/make_claim_matrix.py` resolves each row against the release and writes `outputs/pakemis_claim_matrix.csv`; the tables below print the identifier, the printed and resolved values, the evidence class and the limitation label. The sentence itself, the pointer and the query parameters are in the CSV. A row whose printed value disagrees with the release fails `tests/paper_numbers/test_claim_matrix.py`; nothing is repaired in place.

† printed value outside the DOC_CLAIMS agreement rule at the time of rendering.

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Ahmad, R., 2026b. Satellite-Assisted Nowcasting of Agricultural Greenhouse-Gas Emissions between Census Years. Technical Report. Pakistan Air Quality Initiative. Research note accompanying release 1.0.0rc67; DOI to be assigned.

Table S5: Claim matrix: Abstract.

claim	via	printed	resolved	evidence	limitation
abs_total_L	query	444.96	444.96	closure-exact	incomplete-floor
abs_total_AR6	query	438.19	438.19	closure-exact	
abs_total_S	query	450.20	450.20	structural	incomplete-floor
abs_flat_share	pointer	54.65	54.65	estimated-TE4	monthly-physics
abs_streams_gridded	pointer	199	199	coverage-count	
abs_streams_total	pointer	202	202	coverage-count	
abs_streams_unallocated	pointer	3	3	spatial-S4	
abs_unallocated_share	pointer	0.05	0.06	spatial-S4	
abs_mc_interval	pointer	11.72	11.72	parametric	
abs_mc_down	pointer	10.3	10.3	parametric	
abs_mc_up	pointer	13.1	13.1	parametric	
abs_structural_max	pointer	9.07	9.1	structural	
abs_legacy_control_delta	pointer	17.78	17.8	structural	
abs_completeness_rows	query	138	138	coverage-count	incomplete-floor
abs_completeness_E	query	54	54	coverage-count	incomplete-floor
abs_completeness_NE	query	71	71	coverage-count	incomplete-floor

Table S6: Claim matrix: 1 Introduction.

claim	via	printed	resolved	evidence	limitation
intro_contract_tests	getter	32	32	provenance-count	
intro_enteric_share	pointer	34.1	34.1	closure-exact	
intro_residential_share	pointer	5.4	5.4	closure-exact	
intro_total_L	query	444.96	444.96	closure-exact	incomplete-floor
intro_edgar_magnitude	pointer	87	86	external-comparison	incomplete-floor
intro_edgar_scope	pointer	11	11	external-comparison	incomplete-floor
intro_completeness_rows	query	138	138	coverage-count	
intro_contract_invariants	getter	32	32	provenance-count	
intro_residual_variance	pointer	2.5	2.5	parametric	

Table S7: Claim matrix: 2 Dataset scope and architecture.

claim	via	printed	resolved	evidence	limitation
tab_completeness_rows	query	138	138	coverage-count	incomplete-floor
tab_completeness_codes	pointer	72	72	coverage-count	incomplete-floor

FAO, 2025. FAOSTAT: Emissions totals, crops and livestock, fertilizers by nutrient, and forestry production and trade (wood fuel). Food and Agriculture Organization, Rome. https://www.fao.org/faostat/.

IPCC, 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4: Agriculture, Forestry and Other Land Use. Technical Report. Institute for Global Environmental Strategies, Japan.

IPCC, 2019. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Technical Report. IPCC, Switzerland.

Ministry of Finance, 2024. Pakistan Economic Survey 2023–24, Chapter 2: Agriculture. Technical Report. Government of Pakistan.

Table S8: Claim matrix: 4 Dataset generation.

claim	via	printed	resolved	evidence	limitation
tab_sectors_total	query	444.96	444.96	closure-exact	
fig_sectors_agriculture_share	pointer	50.2	50.2	closure-exact	
tab_categorygas_CH4	query	242.932	242.932	closure-exact	
tab_categorygas_CO2	query	160.944	160.944	closure-exact	
tab_categorygas_N2O	query	41.079	41.079	closure-exact	
fig_series_first_year	query	393.84	393.84	closure-exact	series-basis
fig_series_window_change	query	51.12	51.12	closure-exact	series-basis
fig_seasonality_closure	query	444.96	444.96	closure-exact	
fig_seasonality_flat_share	pointer	54.65	54.65	estimated-TE4	monthly-physics
tab_correlation_national	pointer	11.72	11.72	parametric	
tab_correlation_national_correlated	pointer	11.75	11.75	parametric	
fig_bridge_largest	pointer	9.1	9.1	structural	
tab_dungvariants_difference	query	5.2	5.2	structural	incomplete-floor
res_streams_gridded	pointer	199	199	coverage-count	
res_streams_unallocated	pointer	3	3	spatial-S4	
res_agri_misaligned_crops	pointer	3	3	coverage-count	series-basis
res_agri_calendar_share	pointer	77.37	77.37	estimated-TE4	series-basis
res_agri_basis_change	pointer	-7.6969	-7.6969	structural	series-basis
res_agri_cotton_scope	pointer	1.055	1.055	structural	series-basis
unc_trend_changes_tested	pointer	8	8	parametric	series-basis
unc_trend_changes_resolved	pointer	2	2	parametric	series-basis
unc_trend_common_window	pointer	51.12	51.12	parametric	series-basis
unc_variance_persists	pointer	76.56	76.56	parametric	
unc_variance_residual	pointer	2.48	2.48	parametric	
spatial_adm1_by_year_rows	pointer	7864	7864	coverage-count	
spatial_adm2_by_year_rows	pointer	10350	10350	coverage-count	
spatial_grid_fleet_substituted_fy2015_16	getter	17.66	17.66	spatial-S4	
spatial_stream_years_own_year	getter	795	795	provenance-count	
spatial_stream_years_backcast	pointer	306	306	provenance-count	
spatial_adm2_monthly_by_year_rows	pointer	122616	122616	coverage-count	
spatial_stream_years_timeless	pointer	615	615	provenance-count	

Table S9: Claim matrix: 6 Technical validation.

claim	via	printed	resolved	evidence	limitation
tab_benchmarks_magnitude	pointer	87	86	external-comparison	incomplete-floor

Table S10: Claim matrix: 7 Usage notes and limitations.

claim	via	printed	resolved	evidence	limitation
lim_single_year_period	pointer	FY2023–24	FY2023-24	coverage-count	single-year
lim_floor_edgar_1a2	pointer	44	44	external-comparison	incomplete-floor
lim_floor_industry_gas	getter	5.61	5.61	closure-exact	incomplete-floor
lim_floor_1a2_remaining	pointer	15.85	16	external-comparison	incomplete-floor
lim_floor_chemical_filing	pointer	19.4	19.4	external-comparison	incomplete-floor
lim_floor_commercial	getter	2.4	2.4	structural	incomplete-floor
lim_floor_total_rounded	query	445	445	closure-exact	incomplete-floor
lim_floor_dung_ratio	getter	4.9	4.9	structural	incomplete-floor
lim_floor_config_difference	query	5.25	5.25	structural	incomplete-floor
lim_lulucf_sink	pointer	-4.01	-4.01	structural	lulucf
lim_fugitives_floor	pointer	9.601	9.601	structural	fugitive-envelope
lim_fugitives_top	pointer	23.499	23.499	structural	fugitive-envelope
lim_fugitives_lower	pointer	14.894	14.894	structural	fugitive-envelope
lim_series_first_fy	pointer	FY2015–16	FY2015-16	coverage-count	series-basis
lim_series_calendar_share	pointer	77.37	77.37	estimated-TE4	series-basis
lim_series_ne_years	pointer	no	0	coverage-count	series-basis
lim_firewood_energy_share	getter	38.7	38.7	structural	firewood
lim_dung_energy_share	getter	13.6	13.6	structural	firewood
lim_firewood_uncertainty	pointer	35	35	parametric	firewood
lim_firewood_faostat	pointer	23.89	23.89	structural	firewood
lim_firewood_excluded	pointer	20.81	20.81	structural	firewood
lim_monthly_flat_share	pointer	54.65	54.65	estimated-TE4	monthly-physics
lim_monthly_observed	pointer	37.3	37.3	estimated-TE2	monthly-physics
lim_monthly_physical	pointer	41.0	41.0	estimated-TE4	monthly-physics
lim_monthly_weakly_timed	pointer	21.8	21.8	estimated-TE4	monthly-physics
lim_grade_nine_year	getter	21.76	21.76	estimated-TE4	monthly-grade
lim_grade_reference_year	getter	9.57	9.57	estimated-TE4	monthly-grade
lim_grade_target	getter	15	15	estimated-TE4	monthly-grade
lim_grade_ocac_mass	getter	283.2	283.2	estimated-TE4	monthly-grade
lim_grade_after_ocac	getter	14.68	14.68	estimated-TE4	monthly-grade
lim_managed_soils_moved	getter	95.09	95.11	estimated-TE2	monthly-gap
lim_agriculture_cells	getter	72	72	closure-exact	monthly-gap

Table S11: Claim matrix: 7 Usage notes and limitations (continued).

claim	via	printed	resolved	evidence	limitation
lim_fugitives_moved	getter	93.66	93.66	estimated-TE2	monthly-gap
lim_fugitives_t2_share	getter	43.0	43.0	estimated-TE2	monthly-gap
lim_deposit_sources	pointer	217	217	provenance-count	deposit
lim_deposit_archived	pointer	153	153	provenance-count	deposit
lim_deposit_no_terms	pointer	126	126	provenance-count	deposit
lim_deposit_sources_restated	pointer	217	217	provenance-count	deposit
lim_deposit_files	pointer	1761	1761	provenance-count	deposit

Table S12: Claim matrix: 10 Conclusions.

claim	via	printed	resolved	evidence	limitation
concl_total_L	query	444.96	444.96	closure-exact	incomplete-floor
concl_contract_invariants	getter	32	32	provenance-count	
concl_factor_trap	getter	8.0	8.0	structural	
concl_mc_interval	pointer	11.72	11.72	parametric	
concl_mc_road	pointer	13.3	13.3	parametric	
concl_wastewater_factor	getter	1.6	1.6	structural	
concl_edgar_1a2	pointer	44	44	external-comparison	incomplete-floor
concl_activity_files	pointer	217	217	provenance-count	deposit

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